

RESEARCH ARTICLE

Clinical Diagnostic Accuracy of Parkinson's Disease: Where Do We Stand?

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ABSTRACT: Background: Clinical diagnostic accuracy of Parkinson's disease (PD) remains suboptimal. Changes in disease concept may have improved clinical diagnostic accuracy in the past decade. However, current clinical diagnostic criteria have not been validated against neuropathological confirmation.

Objectives: This study aims to provide up-to-date clinical diagnostic accuracy data and validate current clinical diagnostic criteria for PD against neuropathology.

Methods: A retrospective review of medical records of consecutive patients with parkinsonism from the Queen Square Brain Bank was performed between 2009 and 2019. Clinical diagnosis was documented at early (within 5 years of motor symptom onset) and final stages and categorized by movement disorder experts or regular clinicians. Movement Disorder Society Parkinson's disease (MDS-PD) diagnostic criteria were retrospectively applied. Diagnostic accuracy parameters (sensitivity, specificity, positive/negative predictive value, and accuracy) were calculated using neuropathological diagnosis as the gold standard.

Results: A total of 267 patients (141 PD and 126 non-PD parkinsonism) were included. Clinical diagnostic accuracy was 97.2% for experts, 92.5% for the MDS clinically probable PD criteria, and 90.3% for clinicians. Similar figures were obtained when applied at an early stage (91.5%, 89.5%, and 84.2% diagnostic accuracy, respectively). MDS clinically established early PD criteria demonstrated very high specificity (98.4%) at early stages.

Conclusions: Our results showed an important improvement in PD clinical diagnostic accuracy in clinical practice over the past decade, more marked at early stages of the disease. MDS-PD diagnostic criteria is a valid tool in clinical practice and research for the identification of PD patients showing excellent sensitivity and specificity, although movement disorder experts' diagnosis remains the gold standard PD diagnosis during life. © 2023 International Parkinson and Movement Disorder Society.

Key Words: Parkinson's disease; diagnosis; diagnostic criteria; diagnostic accuracy; clinicopathological study

Despite substantial progress over the past two centuries in the understanding of the disease,¹ no reliable in vivo biomarkers or diagnostic tests are currently available for Parkinson's disease (PD). PD diagnosis remains clinical, with autopsy confirmation as the gold standard. Clinical

diagnostic accuracy varies considerably depending on clinical settings, disease duration, and clinician's expertise.^{2,3} An accurate clinical diagnosis of PD not only is essential for patient care but also has important implications in clinical trials and the development of new treatments and

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health policies. PD clinical diagnosis in the past three decades has remained suboptimal, with figures of diagnostic accuracy about 80% according to a recent meta-analysis.³ Clinical diagnosis is particularly challenging at early stages, and clinicopathological series where diagnostic accuracy was evaluated at an initial or early stage reported an overall diagnostic accuracy of 58%.⁴ Diagnostic accuracy rates slightly increase when diagnosis is provided by movement disorders experts and after a longer follow-up: expert clinician diagnosis has shown to outperform clinical criteria commonly used in clinical practice and has been considered the gold standard diagnosis in life.^{3,5} Therefore, the International Parkinson and Movement Disorder Society (MDS) proposed new Clinical Diagnostic Criteria for Parkinson's Disease (MDS-PD criteria) in 2015 designed to mimic the diagnostic process of expert clinicians with the aim of improving reproducibility and diagnostic accuracy.⁶ The MDS-PD criteria propose two different diagnostic categories according to the level of certainty: clinically probable PD (with a balance between sensitivity and specificity) and clinically established PD (aimed to maximize specificity).⁶ These criteria have been previously validated against expert clinical diagnosis demonstrating a high diagnostic accuracy (overall accuracy: 92.4%, sensitivity: 94.5%, and specificity: 88.5%).⁷ More recently, with the aim of providing a highly specific diagnosis in untreated/early PD patients specially for use in clinical trials, the MDS proposed clinical diagnostic criteria for early PD (clinically established early PD), which have shown an estimated specificity of 95.4% when applied in patients with <5 years of disease duration against expert clinical diagnosis.⁸

Significant advances in the understanding of the clinical presentation of the disease, changes in the disease concept over the past decade, and the publication of new clinical diagnostic criteria may have translated into an improvement in diagnostic accuracy in clinical settings. However, no clinicopathological series have been published in recent years evaluating any potential improvement in clinical diagnostic accuracy and, in addition, the recent diagnostic criteria proposed by the MDS have not been validated against neuropathological diagnosis. Here we report a clinicopathological study of a large recent cohort of consecutive cases with parkinsonian disorders evaluating the diagnostic accuracy against neuropathological diagnosis of (1) the clinical diagnosis of PD in clinical practice and (2) the recent MDS Clinical Diagnostic Criteria for PD to provide neuropathological validation.

Patients and Methods

Study Design and Participants

This retrospective cohort study evaluated consecutive cases presenting with parkinsonism (defined as

bradykinesia in combination with either rigidity or rest tremor)⁶ and brain donation to the Queen Square Brain Bank between 2009 and 2019. The brain donor program and protocol were approved by the London-Central Research Ethics Committee (18/LO/0721), valid informed consent is in place for all cases, and brain tissue is stored for research under a license issued by the Human Tissue Authority (12198).

Clinical Information

Clinical records were systematically reviewed by neurologists with expertise in movement disorders (S.V. and E.P.-F.). All patients had documentation of their illness throughout the entire course from hospital specialists (neurologists and/or geriatricians) in the United Kingdom. Medical records included primary care medical notes, National Hospital for Neurology and Neurosurgery medical records, correspondence from hospital specialists, Queen Square Brain Bank reports, and self-assessment forms. The following diagnostic categories were carefully documented as defined using the available clinical information from the medical records at an early (defined as within 5 years of onset of motor symptoms)^{6,8-10} and final (time of death) stages of the disease.

- MDS-PD criteria (clinically probable PD and clinically established PD categories). Each diagnostic criterion was carefully recorded, and clinical features were considered negative unless documented in the medical records.
- Clinically established early PD (only applied within 5 years of symptom onset).
- Clinical diagnosis: most likely diagnosis at early and final stages as documented by the treating hospital specialist.
- Expert clinical diagnosis: a specific subcategory of clinical diagnosis was created, which included those participants diagnosed by a movement disorder specialist. A movement disorder specialist was defined as a hospital specialist with a dedicated specialist movement disorder clinic in a tertiary care center.

Neuropathological Assessments

Neuropathological assessment of all cases followed standard Queen Square Brain Bank protocols. In summary, brain tissue and, when available, spinal cord were fixed in 10% buffered formalin for 3 weeks, macroscopic changes were documented, and tissue blocks were taken from representative regions. Paraffin-embedded 7- μ m sections were sampled from representative regions, stained using hematoxylin and eosin method, and used for immunohistochemistry with a panel of antibodies recognizing α -synuclein (Abcam, AB1903, 1:10,000), amyloid- β peptide (Dako, Copenhagen, Denmark, 6F/3D, 1:100), and hyperphosphorylated tau (ThermoFisher, anti-human-

PHF-Tau monoclonal antibody, 1:600). When appropriate, differential tau immunohistochemistry was also performed using anti-3-repeat tau-specific (Millipore, 8E6/C11, 1:800) and 4-repeat tau-specific (Millipore, 1E1/A6, 1:4000) antibodies. Neuropathological diagnosis was established according to consensus criteria for PD¹¹ and other parkinsonian disorders, including multiple system atrophy (MSA),^{12,13} progressive supranuclear palsy (PSP),¹⁴ and corticobasal degeneration (CBD),¹⁵ by neuropathologists experienced in the diagnosis of neurodegenerative diseases (T.R. and Z.J.).

Statistical Analysis

Comparisons between those with PD and those with parkinsonian disorders with an alternative neuropathological diagnosis (non-PD parkinsonism) were performed using independent samples *t* test for continuous and χ^2 test for categorical variables as appropriate. Diagnostic performance of each clinical diagnostic category (MDS clinically probable PD, MDS clinically established PD, clinically established early PD, clinical diagnosis, and expert clinical diagnosis), at early (within 5 years of motor symptom onset) and late stages of the disease was compared against the gold standard of neuropathological diagnosis. Diagnostic accuracy parameters were calculated for each of the clinical diagnostic categories (MDS clinically probable PD, MDS clinically established PD, MDS clinically established early PD, clinical diagnosis, and expert clinical diagnosis), including sensitivity (percentage of cases with neuropathology-proven PD that had been correctly diagnosed as PD), specificity (percentage of cases without PD that were diagnosed as not having PD), positive predictive value (percentage of cases correctly diagnosed as PD), and negative predictive value (percentage of cases without neuropathology-proven PD correctly identified as not having PD). Associated 95% confidence interval (CI) was estimated. Comparisons of sensitivity and specificity between each clinical diagnostic category were performed using McNemar's test.^{16,17} To further evaluate the discriminative ability of each diagnostic category, receiver operating characteristic (ROC) curves were plotted using the proportion of true positives against the proportion of false positives, and area under the curve (AUC) was estimated.¹⁸ Interpretation of AUC was based on the following scoring system: 1.0: perfect test, 0.99–0.90: excellent test, 0.89–0.80: good test, 0.79–0.70: fair test, 0.69–0.51: poor test, and 0.50 or lower: fail.¹⁸ ROC curves and AUC values are measures of global diagnostic performance and were provided only for diagnostic categories aiming to provide global diagnostic accuracy. ROC curves and AUC values were not provided for diagnostic categories aiming to maximize specificity (MDS clinically established PD and MDS clinically established

early PD). Significance was set at $P < 0.05$. All statistical analyses were performed using SPSS software, version 23.0 (SPSS Inc., Chicago, IL).

Results

During the study period, 271 patients were identified presenting clinically with parkinsonism. Four patients were excluded due to insufficient clinical information in the medical records regarding disease progression (Fig. 1). A total of 267 patients with parkinsonism were included in this study (57.7% men; age at onset: 61.2 ± 11.1 years; disease duration: 12.4 ± 8.6 years; age at death: 74.0 ± 8.7 years). Full demographic and clinical details are presented in Table 1. Final clinical diagnosis was made by movement disorder experts in 180 patients (67.4%). During follow-up, 40 patients (15%) had a change in the initial clinical diagnosis, of which 8 patients (5.7%) were pathologically confirmed PD and 32 patients (25.4%) were non-PD.

Based on neuropathological diagnosis, 141 patients (52.8%) had PD and 126 patients (47.1%) had non-PD parkinsonism, with neuropathological diagnosis, including MSA in 73 (27.3%), PSP in 44 (16.5%), CBD in 8 (3%), and vascular parkinsonism in 1 (0.4%) (Fig. 1). Table S1 provides additional details on final clinical diagnosis in 267 patients along with their final neuropathological diagnosis. PD patients had an earlier onset (63.4 vs. 59.3, $P = 0.007$), had longer disease duration (17.5 vs. 6.7, $P < 0.005$), and were older at time of death (77.3 vs. 70.4, $P = 0.002$). Sex and disease duration to initial and final diagnoses were similar between groups (Table 1).

Diagnostic Performance at the Early Stage

At the early stage, within 5 years of disease onset, of the 141 PD patients (Tables 2 and 3), a clinical diagnosis of PD was made in 116 (sensitivity: 82.3%), whereas 109 of 126 non-PD were accurately diagnosed as non-PD (specificity: 86.5%; overall accuracy: 84.2%). When the clinical diagnosis was made by movement disorder experts (142 patients: 91 PD and 51 non-PD parkinsonism) in the first 5 years, the overall accuracy improved to 91.5%: of 91 PD patients, 85 were correctly diagnosed as PD (sensitivity: 93.4%), and of 51 non-PD patients, 45 were correctly diagnosed as non-PD (specificity: 88.2%). When applying the MDS-PD criteria at an early stage, 87.9% were diagnosed with PD, as either clinically established PD ($n = 82$, 58.2%) or clinically probable PD ($n = 42$, 29.7%), and 91.3% of non-PD patients were identified as non-PD by the MDS criteria. The overall diagnostic accuracy for clinically probable PD was 89.5%, with a sensitivity of 87.9% and a specificity of 91.3%. MDS clinically established PD had a very high specificity of

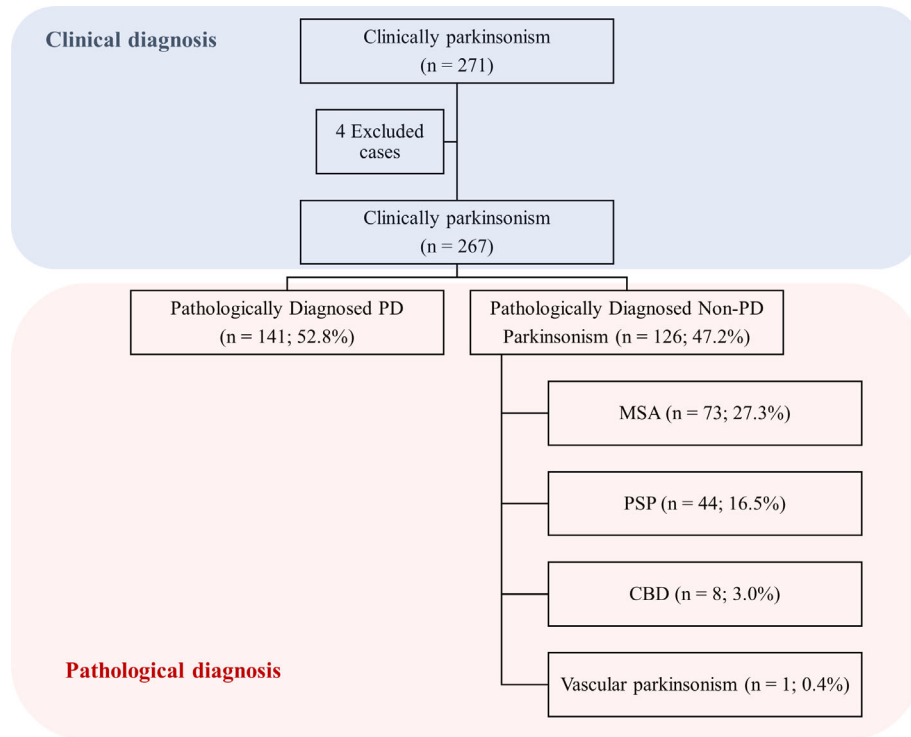


FIG. 1. Study design. PD, Parkinson's disease; MSA, multiple system atrophy; PSP, progressive supranuclear palsy; CBD, corticobasal degeneration. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/mds.23117)]

99.2% although at the expense of modest sensitivity. Similarly, 98.4% of non-PD patients were correctly identified by the clinically established early criteria

showing excellent specificity although only 68.8% of PD patients met these criteria (see Tables 2 and 3 for further details and 95% CI).

TABLE 1 Demographics and clinical characteristics of 267 patients with parkinsonism by pathological diagnosis

	All cases (N = 267)	PD (N = 141)	Non-PD (N = 126)	P-value
Gender, male, N (%)	154 (57.7%)	86 (61.4%)	68 (54.4%)	0.247
Age at death (y)	74.0 (8.7)	77.3 (6.7)	70.4 (9.2)	0.002
Age at disease onset (y)	61.2 (11.1)	59.3 (12.0)	63.4 (9.6)	0.007
Disease duration (y)	12.4 (8.6)	17.5 (8.9)	6.7 (2.8)	<0.001
Disease duration to early clinical diagnosis (y)	1.4 (1.7)	1.3 (1.7)	1.8 (1.7)	0.508
Disease duration to final diagnosis (y)	2.6 (2.3)	2.0 (2.3)	3.3 (2.2)	0.622
Cases where clinical diagnosis was revised after 5 years	40 (15%)	8 (5.7%)	32 (25.4%)	<0.001
Cases where early diagnosis was determined by experts	142 (53.2%)	91 (64.5%)	51 (40.5%)	0.05
Cases where final diagnosis was determined by experts	180 (67.4%)	92 (65.2%)	88 (69.8%)	0.805
Final clinical diagnosis				<0.001
Parkinson's disease	119 (44.5%)	117 (83.3%)	2 (1.6%)	
Multiple system atrophy	80 (30%)	16 (11.3%)	64 (50.8%)	
Progressive supranuclear palsy	56 (21%)	7 (5%)	49 (38.9%)	
Corticobasal syndrome	9 (3.4%)	0	9 (7.1%)	
Other parkinsonism	3 (0.1%)	1 (0.7%)	2 (1.6%)	

Note: Data are presented as number (%) for categorical variables or mean (SD) for continuous variables. P-values correspond to χ^2 test for categorical variables and independent samples *t* test for continuous variables.

Abbreviations: PD, Parkinson's disease; SD, standard deviation.

TABLE 2 Overall diagnostic performance for Parkinson's disease in 267 patients with clinical parkinsonism

Clinical diagnosis/ diagnostic criteria	Pathology-confirmed PD (N = 141)		Pathology-confirmed non-PD parkinsonism (N = 126)	
	Clinical diagnosis PD	Clinical diagnosis non-PD	Clinical diagnosis PD	Clinical diagnosis non-PD
Early stage (<5 years)				
Clinical diagnosis	116 (82.3)	25 (17.7)	17 (13.5)	109 (86.5)
Expert clinical diagnosis ^a	85 (93.4)	6 (6.6)	6 (11.8)	45 (88.2)
MDS clinically established early PD	97 (68.8)	44 (31.2)	2 (1.6)	124 (98.4)
MDS clinically probable PD	124 (87.9)	17 (12.1)	11 (8.7)	115 (91.3)
MDS clinically established PD	82 (58.2)	59 (41.8)	1 (0.8)	125 (99.2)
Final stage				
Clinical diagnosis	117 (83)	24 (17.1)	2 (1.6)	124 (98.4)
Expert clinical diagnosis ^a	87 (94.6)	5 (5.4)	1 (1.1)	87 (98.8)
MDS clinically probable PD	124 (87.8)	17 (12.1)	3 (2.4)	123 (97.6)
MDS clinically established PD	89 (63.1)	52 (36.9)	1 (0.8)	125 (99.2)

Note: Data are presented as number (%).

Abbreviations: PD, Parkinson's disease; MDS, International Parkinson and Movement Disorder Society.

^aExpert clinical diagnosis was made in 142 patients at the early stage and 180 patients at the final stage.

When diagnostic accuracy parameters were compared across diagnostic categories at the early stage, MDS probable PD criteria had both significantly higher sensitivity ($P < 0.001$) and specificity ($P = 0.02$) compared to clinical diagnosis but lower sensitivity compared to

clinical diagnosis made by experts ($P < 0.001$); there was no significant difference in specificity between MDS probable PD criteria and expert clinical diagnosis ($P = 0.063$). As expected, MDS clinically established PD criteria and MDS clinically established early PD

TABLE 3 Sensitivity, specificity, predictive values, and accuracy of different diagnostic categories (and by type of clinician) for the diagnosis of Parkinson's disease at the early stage and after adequate follow-up

	Sensitivity	Specificity	PPV	NPV	Accuracy
Early stage <5 years					
Clinical diagnosis	82.3 (74.9–88.2)	86.5 (79.3–91.9)	87.2 (80.3–92.4)	81.3 (73.7–87.5)	84.3 (79.3–88.4)
Expert clinical diagnosis	93.4 (86.2–97.5)	88.2 (76.1–95.6)	93.4 (86.2–97.5)	88.2 (76.1–95.6)	91.5 (85.7–95.6)
MDS clinically established early PD	68.8 (60.5–76.3)	98.4 (94.4–99.8)	98.0 (92.9–99.8)	73.8 (66.5–80.3)	NA ^a
MDS clinically probable PD	87.9 (81.4–92.8)	91.3 (84.9–95.6)	91.9 (85.9–95.9)	87.0 (80.2–92.3)	89.5 (85.2–92.9)
MDS clinically established PD	58.2 (49.6–66.4)	99.2 (95.7–100.0)	98.9 (93.5–100.0)	67.9 (60.7–74.6)	NA ^a
Final stage					
Clinical diagnosis	83.0 (75.7–88.8)	98.4 (94.4–99.8)	98.3 (94.1–99.8)	83.8 (76.8–89.3)	90.3 (86.1–93.5)
Expert clinical diagnosis	94.6 (87.8–98.2)	98.9 (93.8–100.0)	98.9 (93.8–100.0)	94.6 (87.8–98.2)	96.7 (92.9–98.8)
MDS clinically probable PD	87.9 (81.4–92.8)	97.6 (93.2–99.5)	97.6 (93.3–99.5)	87.9 (81.3–92.8)	92.5 (88.7–95.4)
MDS clinically established PD	63.1 (54.6–71.1)	99.2 (95.7–100.0)	98.9 (94.0–100.0)	70.6 (63.3–77.2)	NA ^a

Note: Data are presented as percentage (95% confidence interval).

Abbreviations: PPV, positive predictive value; NPV, negative predictive value; MDS, International Parkinson and Movement Disorder Society; PD, Parkinson's disease; NA, not applicable.

^aAs MDS clinically established categories were designed to maximize specificity, a measure of global diagnostic accuracy is not considered appropriate.

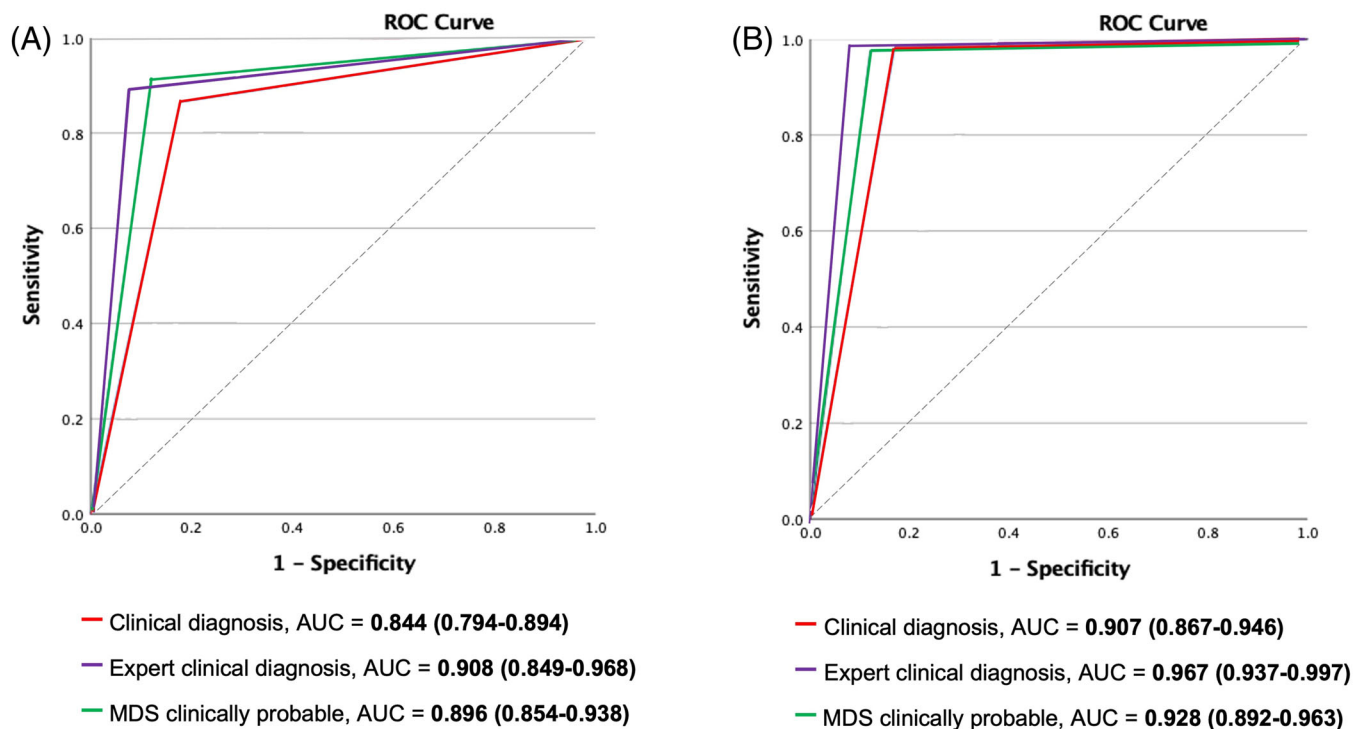


FIG. 2. ROC curves of each diagnostic category for Parkinson's disease at (A) early and (B) final stages. AUC, area under the ROC curve; ROC, receiver operating characteristic. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/mds.23117)]

criteria showed significantly higher specificity at an early stage compared to both clinical diagnosis and expert clinical diagnosis ($P < 0.001$ for all comparisons).

To visually investigate global diagnostic accuracy for each diagnostic category, ROC curves were plotted (Fig. 2A). AUC for clinical diagnosis of PD was 0.844 (“good”¹⁸), for expert clinical diagnosis was 0.908 (“excellent”), and for MDS clinically probable PD was 0.896 (“good”) (see Fig. 2A for 95% CI).

Diagnostic Performance at the Final Stage

Clinical diagnosis was reviewed in 40 patients (15%), improving the diagnostic accuracy at the final stage; 83% of PD patients and 98.4% of non-PD patients were diagnosed correctly at the final stage as PD and non-PD, respectively, improving the overall accuracy to 90.3% (Tables 2 and 3). Clinical diagnostic accuracy was higher in those 180 patients with a final clinical diagnosis made by movement disorder experts, with an overall accuracy of 96.7.

Throughout the course of the disease, 4 patients initially not meeting MDS diagnostic criteria for PD at early stage developed further supportive features and were categorized as clinically probable PD by the MDS diagnostic criteria at the final stage. Similarly, 6 patients with clinically probable PD at early stage became clinically established PD at the final stage after developing further

supporting clinical features although red flags emerged in 2 patients initially classified as clinically probable PD and did not meet MDS diagnostic criteria. No changes occurred in those initially diagnosed as clinically established PD by the MDS criteria, resulting in the overall percentage agreement of MDS-PD criteria comparing categorization at earlier stage to final stage at 94.3% (Table S2).

Similarly, MDS diagnostic criteria accuracy improved when applied at the final stage of the disease. Of the PD patients, 24.7% and 63.1% met MDS criteria for probable and clinically established PD, respectively, and 97.6% of non-PD were identified as non-PD using the criteria. Combining all patients, the overall accuracy for MDS clinically probable PD was 92.5%. In addition, 63.1% of PD patients met MDS criteria for clinically established PD, whereas only 0.8% of non-PD patients met clinically established criteria. Therefore, in the final stage, MDS probable PD criteria demonstrated a sensitivity of 87.9% significantly higher than clinical diagnosis (sensitivity: 83%, $P < 0.001$), but still slightly lower than expert clinical diagnosis (sensitivity: 94.6%, $P < 0.001$), and a specificity of 97.6%, similar to clinical diagnosis (specificity: 98.4%, $P = 0.087$) but slightly lower than expert clinical diagnosis (specificity: 98.9%, $P = 0.034$). As expected, MDS clinically established criteria had a very high specificity for diagnosis of PD at 99.2%, significantly higher than clinical

diagnosis ($P < 0.001$), but far lower sensitivity at 63.1% compared to other diagnostic categories ($P < 0.001$). See Tables 2 and 3 for further details and 95% CI.

The ROC curve for each diagnostic category at the final stage is represented in Figure 2B. ROC curves for clinical diagnosis, expert diagnosis, and MDS clinically probable categories showed excellent AUC values (0.907, 0.967, and 0.928, respectively).

Discussion

In this clinicopathological study of patients presenting with parkinsonism, the overall PD diagnostic accuracy among clinicians was 84.2%, with the sensitivity, specificity, and predicted values that clearly exceeded the 80% target threshold even early in the course of the disease. All diagnostic parameters, as expected, improved at final diagnosis after adequate follow-up, in which the overall diagnostic accuracy of the clinical diagnosis reached 90.3%. The most frequent misdiagnoses were MSA and PSP, in line with previous autopsy series.^{3,19,20}

The clinical diagnostic accuracy parameters from our current study have improved compared to similar historical clinicopathological series over the past three decades.^{21,22} A recent meta-analysis,³ including 28 studies from 1988 to 2014, demonstrated that the overall accuracy of PD clinical diagnosis was 80.6%. Most recent clinicopathological series published in the early 21st century failed to demonstrate a clear trend of improvement in diagnostic performance compared to studies from the 20th century, which may be explained by the historical focus of the diagnosis on the motor aspects of the disease. Fundamental changes in the disease concepts over the past decade, the wide recognition of PD as a multisystem disorder associated with multiple nonmotor features, the improvement in the diagnostic tests available in clinical practice for the differential diagnosis, and the publication of updated clinical diagnostic criteria reflecting the advances in the understanding of the disease may all have contributed to an improvement in the diagnostic assessment in clinical practice.

Our study also provides pathological validation for the currently most widely accepted clinical diagnostic criteria for PD, and we compared their diagnostic accuracy against the clinical diagnosis made by experts and nonexperts in movement disorders at the early and final stages of the disease. The MDS-PD diagnostic criteria had been previously validated using expert clinical diagnosis as the gold standard⁷ with a diagnostic accuracy of 92.4%, a sensitivity of 94.5%, and a specificity of 88.5%. We found that on retrospectively applying MDS-PD diagnostic criteria in a cohort of pathology-

proven patients with PD and non-PD parkinsonism the diagnostic performance of clinically probable PD reached similar diagnostic performance with an overall accuracy of 92.5% although sensitivity and specificity showed a slightly inverted balance (sensitivity: 87.9%, specificity: 97.6%). Overall, these figures demonstrate excellent diagnostic performance and a significant improvement compared to historical accuracy data and current clinical practice. These criteria were designed to mimic the diagnostic process of movement disorder experts, which is considered the *in vivo* gold standard diagnosis, and, when these data were compared with expert clinical diagnosis in our study, no significant differences were found. Diagnostic accuracy rates among experts in our study are similar to a previous clinicopathological series published in 2002 from the Queen Square Brain Bank with a similar design.²³ Diagnostic accuracy among experts at final stages in our series (reached in our cohort at 2.6 years after onset) showed the best diagnostic performance with an overall accuracy of 96.7%, which provides an excellent *in vivo* gold standard. We also found the diagnostic PD categories to be stable; patients in both established and probable groups retained a PD categorization of about 98% at final stage (Table S2).

Similarly, applying clinically established early PD criteria at early stages of the disease in our cohort provided excellent specificity (98.4%) with sufficient sensitivity (68.8%). These figures are also in line with a recent validation study (specificity: 95.4%, sensitivity: 69.8%) using expert clinical diagnosis as the gold standard.⁸ It should be noted that these criteria were designed for the recruitment of participants in research at early stages of the disease, where a very high specificity in the diagnosis is recommended, and not to provide global diagnostic accuracy.

Another important finding of our study is the diagnostic accuracy rates at early stages (<5 years from onset). Although it is expected that diagnostic accuracy will improve as the disease progresses and this has been well documented in previous studies,^{3,4,20,24} our current data showed that diagnostic accuracy performance over time was only marginally better for both, clinicians and MDS-PD diagnostic criteria. Overall diagnostic accuracy at early stages in our study was 84.2%, 89.5%, and 91.5% for clinical diagnosis, MDS clinically probable PD criteria, and expert diagnosis, respectively. These figures are in clear contrast with previous clinicopathological series that reported a poor diagnostic accuracy of 58%. These data suggest that clinical diagnostic accuracy has improved over the past decade and that an accurate diagnosis is reached at earlier stages of the disease. The most common misdiagnosis in clinical practice is MSA, whereas only very few patients with non-PD parkinsonism were misclassified clinically as PD. This tendency to clinically overdiagnose MSA

suggests that clinicians may fail to recognize atypical presentations of PD, particularly those presenting with overlapping symptoms early in the disease course (early autonomic dysfunction).

Although reliable diagnostic biomarkers are still lacking, seeding amplification assays (protein misfolding cyclic amplification and real-time quaking-induced conversion) for the detection of misfolded α -synuclein in the cerebrospinal fluid and skin biopsies have shown very promising results.^{25,26} Several studies have demonstrated sensitivity and specificity figures greater than 90% to detect Lewy body disorders at early stages and differentiate PD from MSA,²⁷ and the implementation of seeding assays in clinical practice is expected to help clinicians in the accurate diagnosis of PD, even at prodromal stages.

The main strength of the study is the presence of neuropathological confirmation of the diagnosis in a large cohort of PD patients with detailed clinical information. Our study also included a large group of patients with other parkinsonian disorders that are most commonly found in the differential diagnosis of PD. Detailed clinical information throughout the whole disease course was available for the patients included in the study, which allowed their diagnostic characterization at different stages of the disease and the application of the MDS clinical diagnostic criteria. Similarly, our cohort included patients evaluated in secondary and tertiary care, which allowed a good representation of clinical practice at different levels of care, including experts and nonexperts, in movement disorders. This could offset the between-study heterogeneity commonly occurring in systematic reviews and meta-analyses.^{28,29}

Our study has several limitations. Inherently to clinicopathological studies, brain bank cases may be biased toward more severe or atypical presentations, which should be considered in the interpretation and generalization of the results. Although this could potentially result in higher clinical misdiagnosis rates, we attempted to minimize this potential bias by including a control group of non-PD parkinsonism. Our study mainly relied on the documentation by treating clinicians on medical records during routine clinical practice. Patients included in the study have detailed documentation, and regular evaluations of their disease process and efforts were made to exclude patients with insufficient clinical information. However, due to the assessment by different clinicians without clear methodological homogeneity and the retrospective nature of the study, we acknowledge that some symptoms and clinical features, particularly those with limited clinical relevance, may potentially account for some limitations in the documentation of clinical information, particularly involving some clinical features of limited clinical relevance. Any additional investigations in our cases were requested based on clinical needs, and we note

that some supportive features of the MDS-PD diagnostic criteria require investigations of limited availability in clinical practice (eg, olfactory testing or cardiac metaiodobenzylguanidine scintigraphy). We acknowledge that this may have had an impact on the retrospective application of the MDS diagnostic criteria and their diagnostic accuracy.

In conclusion, our clinicopathological study demonstrated an improvement in the clinical diagnostic accuracy over the past decade, even at early stages of the disease, which may be explained by the incorporation of advances in the understanding of the disease into the diagnostic process in clinical practice. Validation of the current MDS-PD diagnostic criteria showed excellent diagnostic performance, with the MDS clinically probable PD category showing a good balance between sensitivity and specificity and diagnostic accuracy similar to the clinical diagnosis performed by movement disorder experts. Similarly, MDS clinically established early PD criteria are highly specific allowing the identification of de novo PD cases with a high level of certainty for targeted intervention trials at early stages of the disease. These findings represent important improvements in the diagnosis of PD in clinical practice and clinical research and have important implications until a reliable biomarker becomes available. ■

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Data Availability Statement

Data are not provided in the article because of space limitations but may be shared (anonymized) at the appropriate request of any qualified investigator, in compliance with regulatory ethical approvals, for purposes of replicating procedures and results.

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Supporting Data

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